

Airline Reservation System – High-Level Design

1. Overview and scope

The backend for a single airline. A back office defines flight schedules; customers search flights, view seat maps, book seats and cancel bookings. Bookings are accepted for flights up to 365 days ahead; the window is configuration (`airline.booking-window-days`).

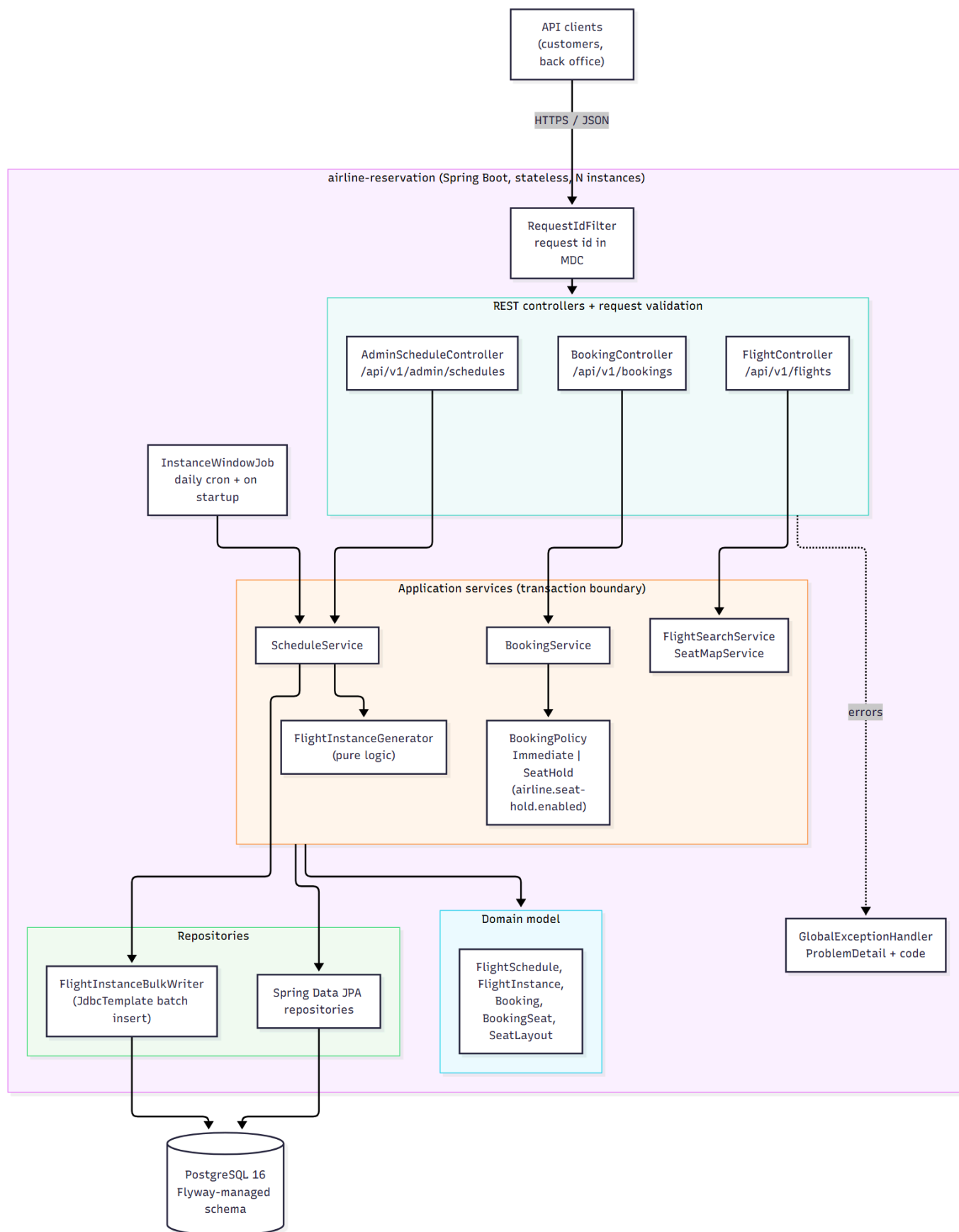
In scope: schedule creation (admin), flight instance generation, flight search, seat map, booking, booking lookup, cancellation. An optional **seat hold** (book now, confirm later) can be enabled by configuration. It is off by default, and with it off the system behaves exactly as the brief describes.

Out of scope: authentication and authorisation, payments and refunds, airport and aircraft CRUD, schedule edits or deletes, time-zone conversion, passenger validation, multi-leg itineraries, pricing and fare classes, a scheduled clean-up job for holds.

How to build, run and try it: [README](#). Detailed design: [LLD](#). Decision records with rejected alternatives: [docs/adr](#) .

2. Architecture

One deployable Spring Boot service (a layered monolith, packaged by feature) in front of one PostgreSQL database. The service holds no state between requests, so any number of instances can run behind a load balancer. Every correctness guarantee is enforced by the database, not by in-memory locks.



Source: [diagrams/architecture.mmd](#)

Layer	Responsibility	Rules
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REST controllers	HTTP mapping, request-shape validation (Bean Validation), status codes	No business logic, never transactional, never touch repositories
Application services	One public method per use case; the only transaction boundary	@Transactional here and nowhere else; readOnly for queries
Domain model	JPA entities with behaviour (reserve, release, cancel, confirm, expireHold), the SeatLayout value object, the BookingStatus state machine	No public status setters; an invalid transition throws IllegalStateException (a bug, never a user error)
Repositories	Persistence only: Spring Data JPA, plus one JDBC class for the bulk instance insert	No business rules
PostgreSQL	Source of truth; constraints are the last line of defence	Schema owned by Flyway; Hibernate only validates

3. Components

Component	Contents	Purpose
airport	Airport, repository	Seeded reference data (IATA code, name, city, country)
aircraft	Aircraft, SeatLayout, repository	Seeded aircraft with a fixed seat configuration (rows × seat letters)
schedule	AdminScheduleController, ScheduleService, FlightSchedule, FlightInstanceGenerator, InstanceWindowJob	Back-office creation and lookup of schedules (templates); turning schedules into flight instances
flight	FlightController, FlightSearchService, SeatMapService, FlightInstance, FlightInstanceBulkWriter	Materialised flight instances, search and seat map
booking	BookingController, BookingService, BookingPolicy (+2 implementations), BookingRequestValidator, PnrGenerator, Booking, BookingSeat	Booking, lookup, confirmation, cancellation
common	ClockConfig, AirlineProperties, GlobalExceptionHandler, RequestIdFilter	Configuration, time, error contract, request-id logging

`InstanceWindowJob` runs inside the service: a daily cron job, plus a run at startup, that keeps every schedule's instances generated 365 days ahead.

Feature dependencies point one way, with no cycles: `booking` → `flight` → `aircraft`, `airport` and `schedule` → `flight`, `aircraft`, `airport`; every feature may use `common`. The seat map and search read booking-seat rows with SQL in `flight`'s persistence layer, so `flight` does not depend on `booking` code.

Data model

AIRPORT			
varchar(3)	code	PK	IATA, ^[A-Z]{3}\$
varchar(150)	name		
varchar(100)	city		
varchar(100)	country		

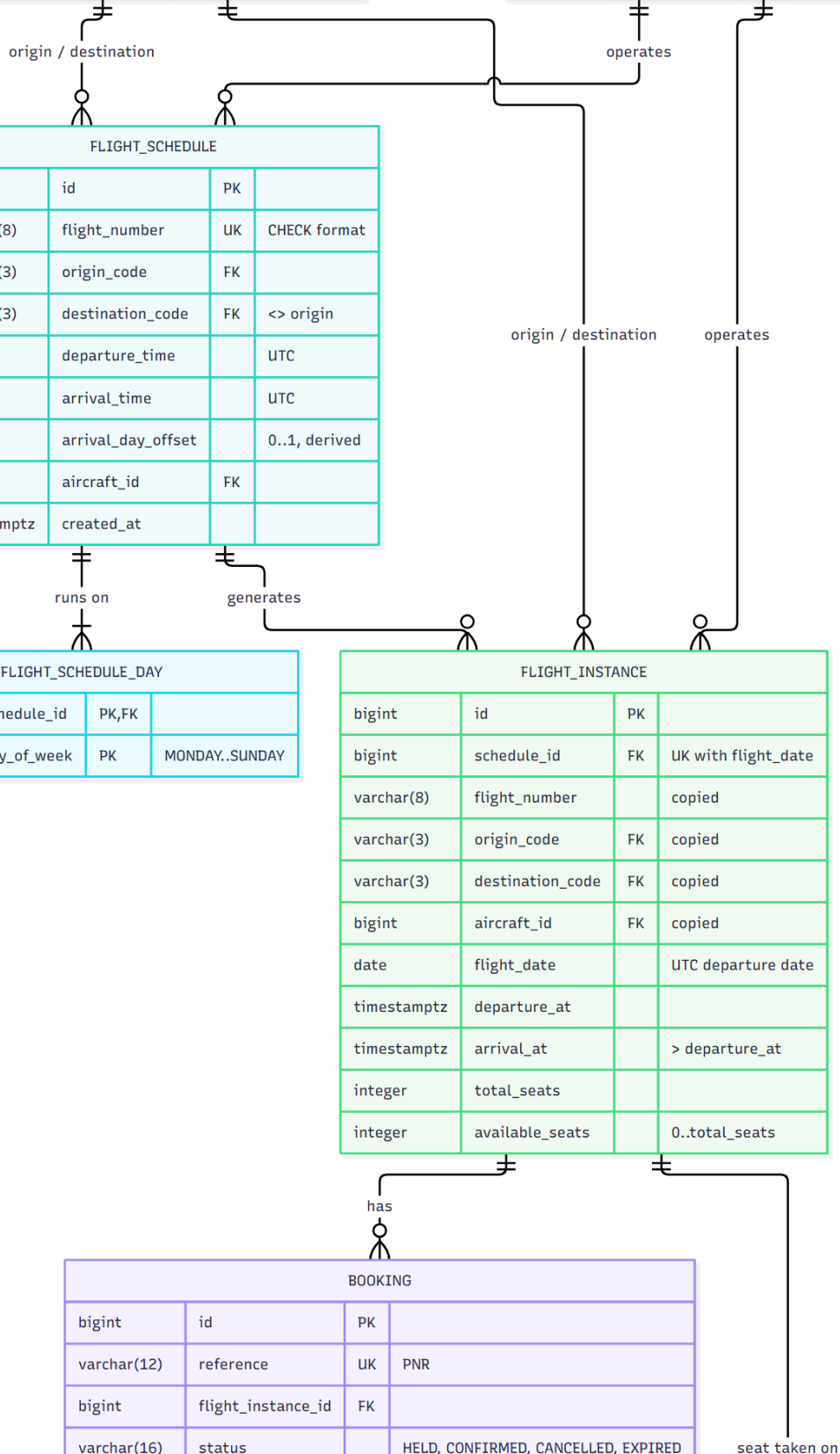
AIRCRAFT			
bigint	id	PK	
varchar(20)	code	UK	registration
varchar(50)	aircraft_type		e.g. A320
integer	row_count		1..99
varchar(12)	seat_letters		e.g. ABCDEF

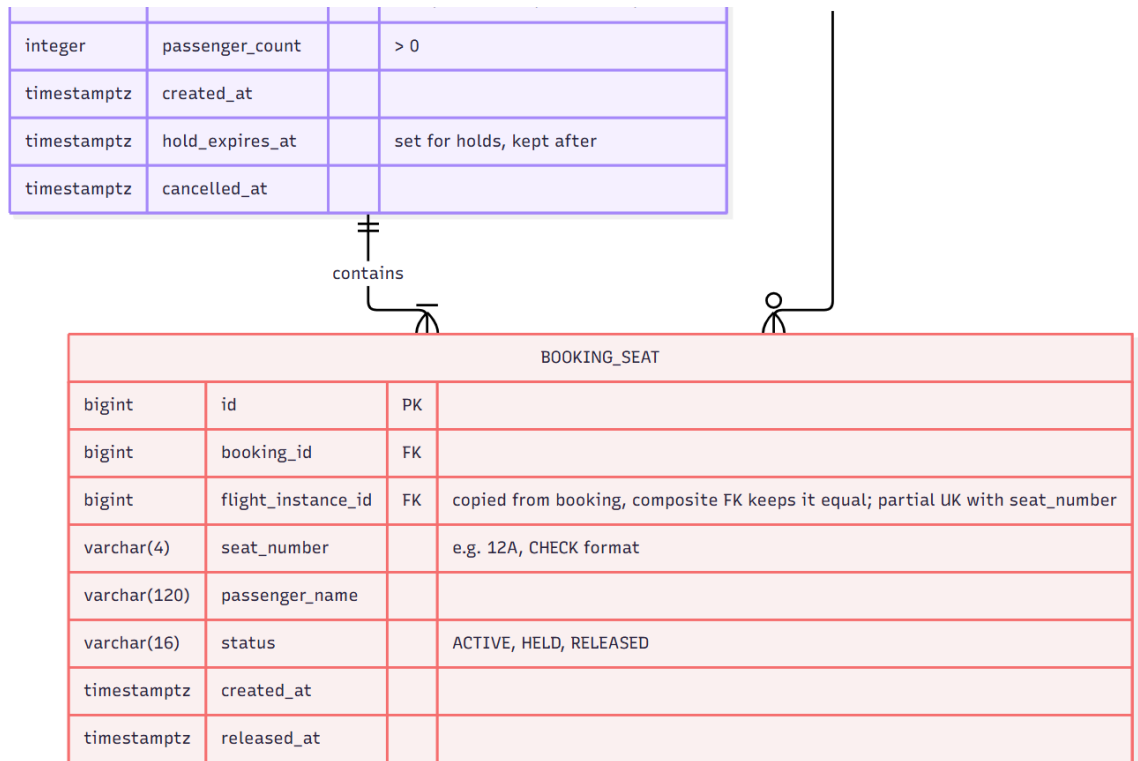
FLIGHT_SCHEDULE			
bigint	id	PK	
varchar(8)	flight_number	UK	CHECK format
varchar(3)	origin_code	FK	
varchar(3)	destination_code	FK	<> origin
time	departure_time		UTC
time	arrival_time		UTC
integer	arrival_day_offset		0..1, derived
bigint	aircraft_id	FK	
timestampz	created_at		

FLIGHT_SCHEDULE_DAY			
bigint	schedule_id	PK,FK	
varchar(9)	day_of_week	PK	MONDAY..SUNDAY

FLIGHT_INSTANCE			
bigint	id	PK	
bigint	schedule_id	FK	UK with flight_date
varchar(8)	flight_number		copied
varchar(3)	origin_code	FK	copied
varchar(3)	destination_code	FK	copied
bigint	aircraft_id	FK	copied
date	flight_date		UTC departure date
timestampz	departure_at		
timestampz	arrival_at		> departure_at
integer	total_seats		
integer	available_seats		0..total_seats

BOOKING			
bigint	id	PK	
varchar(12)	reference	UK	PNR
bigint	flight_instance_id	FK	
varchar(16)	status		HELD, CONFIRMED, CANCELLED, EXPIRED





Source: [diagrams/er.mmd](#) . Full DDL with keys, constraints and indexes: [db/schema.sql](#) .

- `flight_schedule` (+ `flight_schedule_day`) is the template; `flight_instance` is one dated, bookable flight. The route, flight number and aircraft are copied onto the instance because schedules never change, so search reads one table.
- `booking` holds the customer-facing reference and status; `booking_seat` holds one row per passenger and seat. Passengers are not a separate table: one name per seat is all the brief requires.
- `booking_seat.flight_instance_id` duplicates the booking's, so that the double-booking index can be declared on one table. A composite foreign key `(booking_id, flight_instance_id) → booking (id, flight_instance_id)` guarantees the copy always equals the booking's flight.

4. Service interactions and request flows

All endpoints live under `/api/v1` . Admin endpoints sit under `/api/v1/admin` .

4.1 Create schedule (back office)

`POST /admin/schedules` → `ScheduleService.createSchedule` , in one transaction: validate (airports exist and differ, aircraft exists, at least one day, flight number unused) → derive the arrival-day offset → insert the schedule → generate instances for [today, today + 365] → bulk insert them → return the schedule with the generated count.

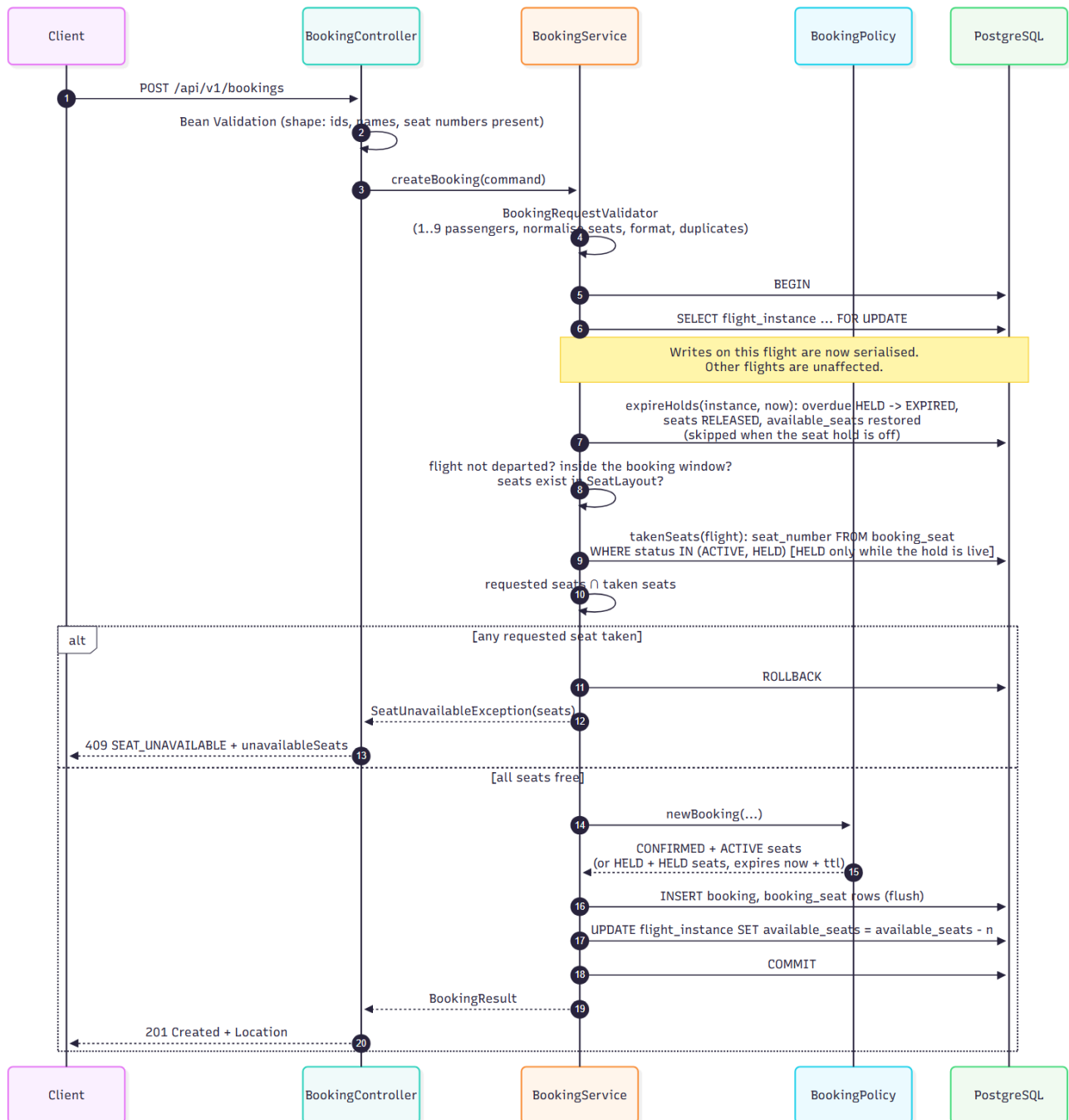
4.2 Search

`GET /flights?origin=&destination=&date=` → `FlightSearchService.search` , read-only: validate the codes and that the date is inside the booking window → check both airports exist → one indexed query on `flight_instance` by (origin, destination, date) where `departure_at > now` , ordered by departure. No day-of-week logic runs at search time: schedules were matched to dates when the instances were generated. `availableSeats` comes from the instance's counter column.

4.3 Seat map

`GET /flights/{id}/seats` → `SeatMapService.getSeatMap` , read-only: load the instance and its aircraft → load the seat numbers that are taken on that instance (one query) → walk the aircraft's `SeatLayout` (row 1..N, then letters in configured order) and mark each seat `BOOKED` or `AVAILABLE` . There is no lock: the map reflects everything committed at the moment of the query.

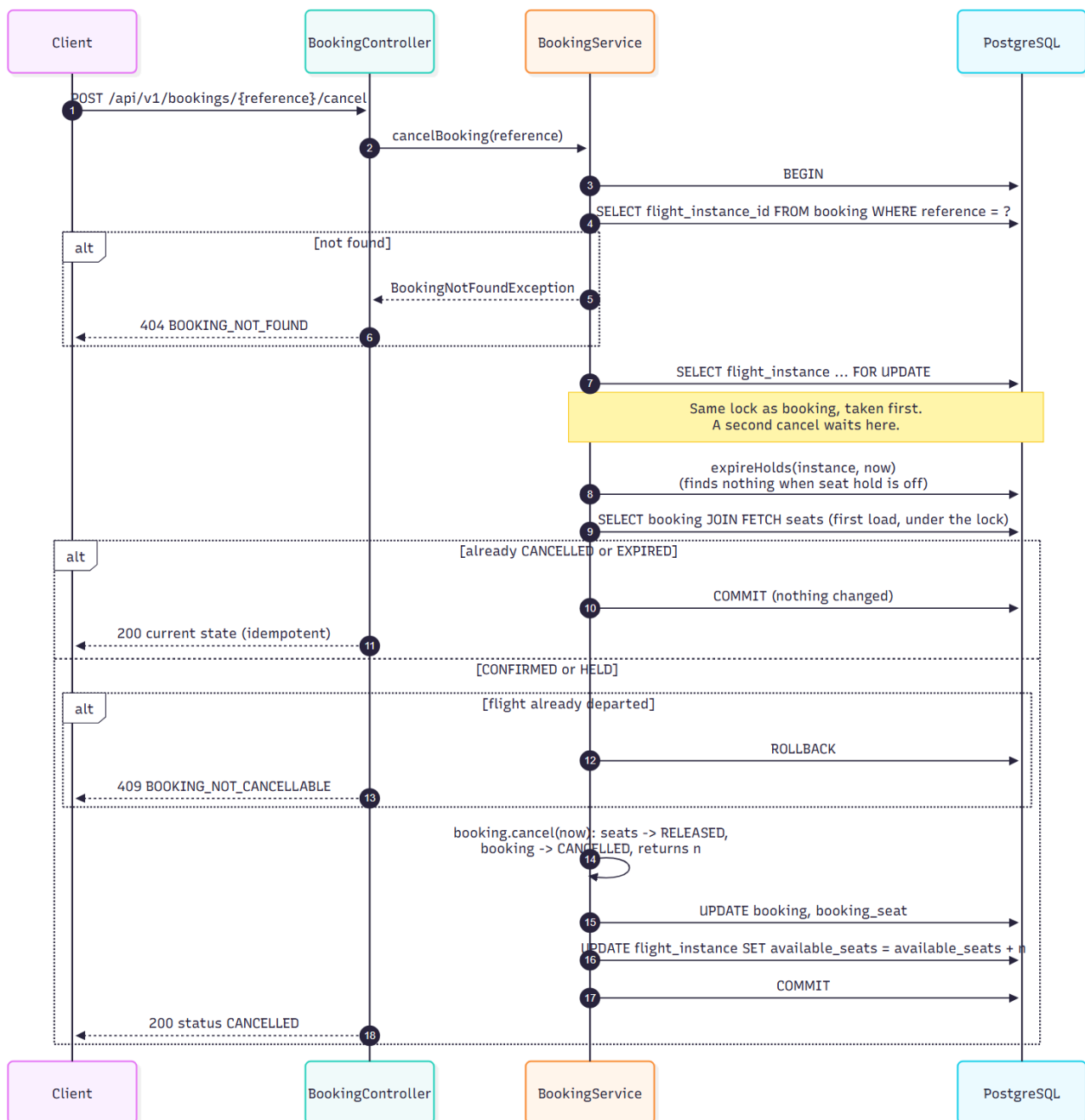
4.4 Booking



Source: [diagrams/booking-sequence.mmd](#)

The booking is all-or-nothing: if any requested seat is taken, nothing is booked and the response lists the taken seats.

4.5 Cancellation



Source: [diagrams/cancellation-sequence.mmd](#)

Cancellation is a soft release. The booking becomes `CANCELLED` and its seats `RELEASED`, so history is kept and the seats are bookable again at once. Cancelling twice returns the same result both times.

4.6 Lookup and confirmation

- `GET /bookings/{reference}` : a read-only load by the customer-facing reference.
- `POST /bookings/{reference}/confirm` (meaningful when the seat hold is on): locks the flight like cancellation does, then moves `HELD` → `CONFIRMED` if the hold has not expired. Confirming a booking that is already `CONFIRMED` returns it unchanged.

5. Flight generation strategy

Decision: materialise one `flight_instance` row per operating date; do not materialise seats.

- When a schedule is created, its instances for [today, today + 365] are generated in the same transaction. For each date in the window whose UTC day of week is one of the schedule's operating days, one row is created with `departure_at / arrival_at` as UTC instants and `available_seats = total_seats = rows × letters`.
- **Rolling window.** The same top-up runs daily (cron, UTC), once at startup, and on demand through `POST /api/v1/admin/instance-window/extend` (e.g. after lengthening the window). It re-generates [today, today + window + 1 day] for every schedule and inserts with `ON CONFLICT (schedule_id, flight_date) DO NOTHING`. The extra day means the newest bookable date already has its instances when the window moves at midnight, before the daily run; search and booking still stop at today + window. The unique constraint makes the job idempotent: a re-run, a run after downtime, or a run on several nodes at once inserts only the missing rows.
- **Seats are derived, not stored.** A seat map is the aircraft's layout minus the taken seats. Only booked seats have rows (`booking_seat`), so storage grows with bookings, not capacity.
- **Overnight flights.** If the arrival time is at or before the departure time, the arrival is on the next calendar day (`arrival_day_offset = 1`). The offset is derived, never supplied by the client.

Why materialise instances: a booking needs a stable row to reference by foreign key and a concrete row to lock, and search becomes a single indexed query. Volume is small: a daily flight is 366–367 rows a year, and a few hundred schedules come to about 100,000 rows a year.

Alternative	Why not
Fully dynamic (compute instances from schedules at query time)	No row to reference or lock; seat availability would need its own keyed storage anyway; search must evaluate every schedule
Lazy hybrid (create the instance on first booking)	Adds a create-if-absent race to the hot booking path; search still has to compute from schedules
Materialise every seat (row per seat per instance)	180–400 rows per instance, almost all never touched; up to ~40 million rows a year at the target scale for no gain

6. Concurrency strategy

Decision: a pessimistic row lock on the flight instance, backed by a partial unique index.

1. **Primary – serialise writes per flight.** Every write transaction (book, confirm, cancel) starts by locking its `flight_instance` row (`SELECT ... FOR UPDATE, JPA PESSIMISTIC_WRITE`). Two requests for the same flight run one after the other; requests for different flights run in parallel. The seat-conflict check, the inserts and the counter update all happen while the lock is held.
2. **Backstop – the database refuses a double booking.** The partial unique index `uq_active_seat on booking_seat (flight_instance_id, seat_number) WHERE status IN ('ACTIVE','HELD')` makes it impossible to store two live claims on one seat, even if application logic were wrong. A violation is mapped to `409 SEAT_UNAVAILABLE`, never a 500.

Isolation level: READ COMMITTED (the PostgreSQL default). Each statement sees the latest committed data. Because the seat-conflict query runs after the lock is granted, it sees every booking committed by the previous lock holder. A stricter level adds serialisation failures and retries without adding safety here.

Lock timeout. Connections set `lock_timeout = 3s`. A request that cannot get the flight lock in time fails fast with `503 LOCK_TIMEOUT` and `Retry-After: 1` instead of queueing indefinitely.

Lock ordering and deadlocks. Every write path takes the flight-instance lock first, before reading or touching booking rows, and a transaction only ever locks one instance. With one lock per transaction there is no cycle, so overlapping multi-seat requests (1A+1B vs 1B+1C) cannot deadlock; one waits and then sees the other's seats as taken.

Why cancellation needs the lock. The booking reference identifies a fixed set of seats, and releasing a seat twice changes nothing; the risk is the availability counter. Two unlocked cancellations would both see `CONFIRMED` and both add the seats back, so search would advertise seats that do not exist. The booking row is loaded only after the flight lock is held: the second cancellation waits, then sees `CANCELLED` and returns it unchanged. Locking the flight first on every path also keeps a single lock order, so cancellation cannot deadlock with booking or hold expiry.

Alternative	Why not
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Optimistic locking (@Version on the instance or booking)	Popular flights are high-contention writes; optimistic versions turn contention into failed requests and client retries. The row lock is held for milliseconds
Unique index only, no lock	Prevents double booking, but cannot keep the availability counter consistent without more machinery, and gives vaguer errors for multi-seat requests
Per-seat row locks	Needs a row per seat (see §5) and multi-seat requests must lock in a fixed order to avoid deadlocks
JVM locks (synchronized, in-memory maps)	Only correct on a single instance; breaks as soon as the service scales out
SERIALIZABLE isolation	Correct, but shifts the problem to retry loops on serialisation failures

7. Seat availability

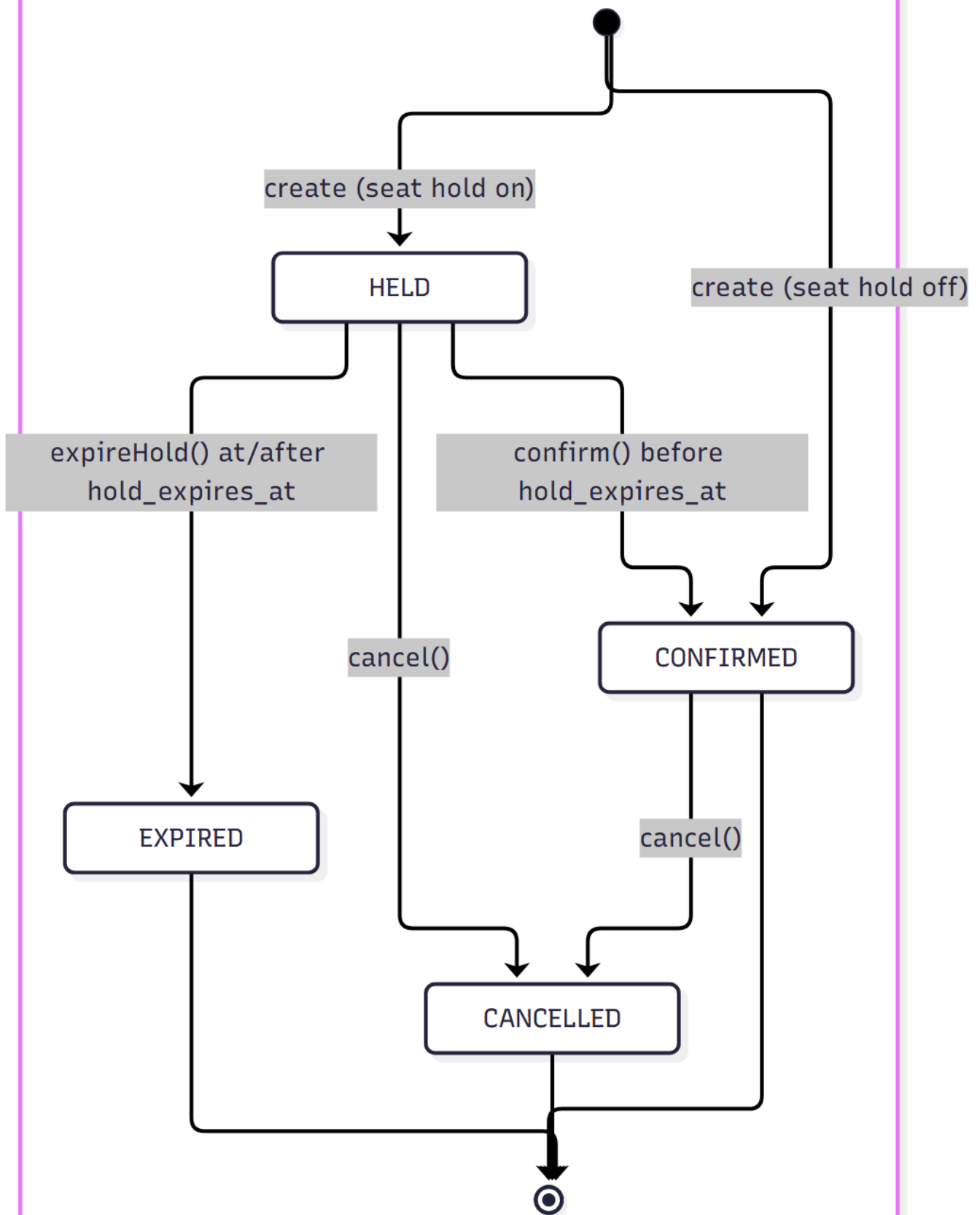
`flight_instance.available_seats` is a counter maintained inside the transaction that holds the flight lock: decremented by a booking (or hold), incremented by a cancellation (or expiry). Search reads it directly, with no aggregation. A `CHECK (available_seats BETWEEN 0 AND total_seats)` bounds it. The invariant, asserted in tests, is:

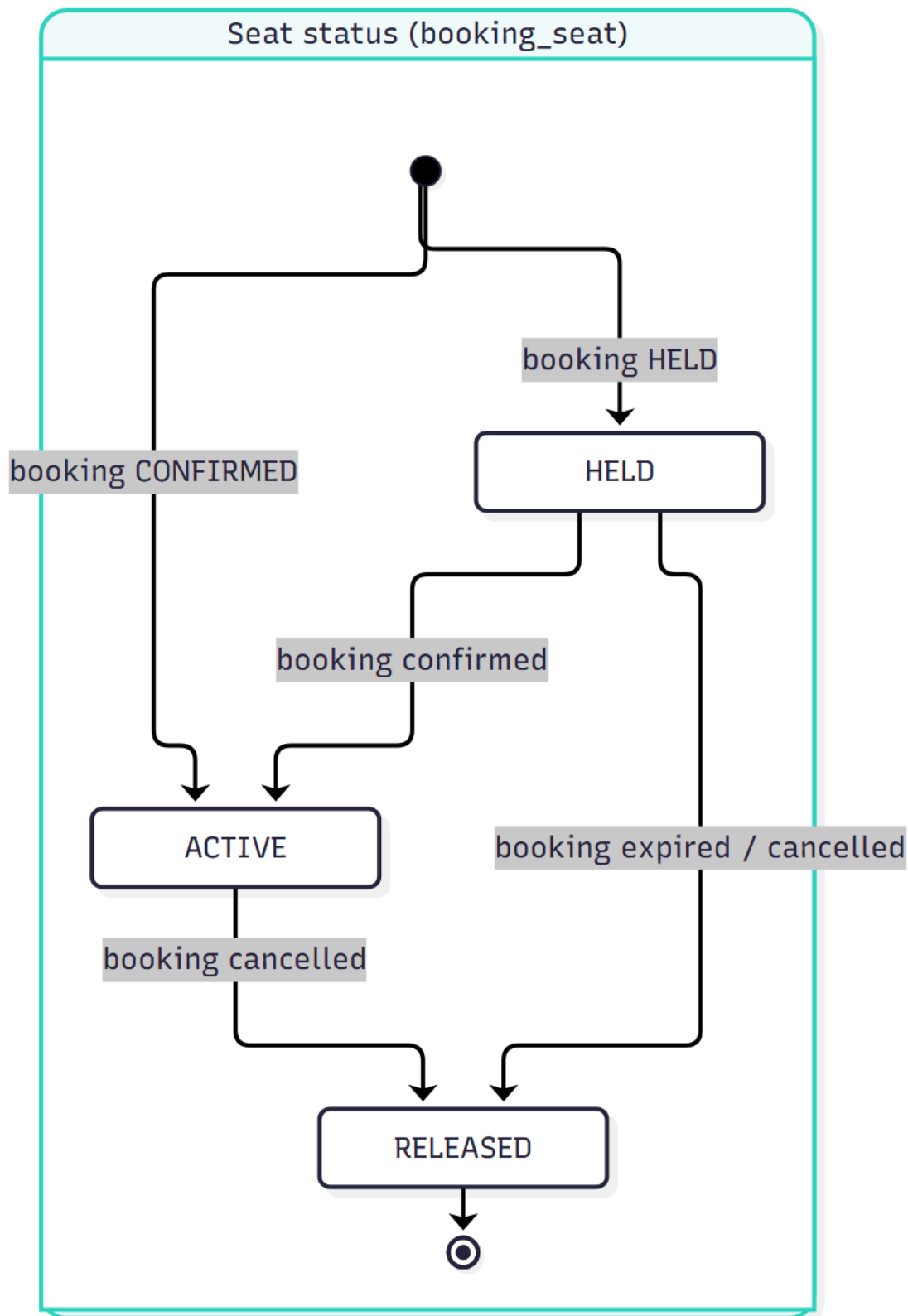
```
available_seats = total_seats - count(booking_seat rows with status ACTIVE or HELD)
```

8. Seat hold (optional, off by default)

Enabled with `airline.seat-hold.enabled=true`; hold length `airline.seat-hold.ttl` (default 10 minutes). The choice is a `Strategy (BookingPolicy)` selected once from configuration: `ImmediateConfirmationPolicy` (default) or `SeatHoldPolicy`.

Booking status





Source: [diagrams/booking-state.mmd](#)

- **Create:** the booking is HELD, its seats HELD, with hold_expires_at = now + ttl. A held seat is protected by the same unique index and counted out of available_seats exactly like a confirmed one.
- **Confirm:** HELD → CONFIRMED, seats → ACTIVE, only if hold_expires_at > now; otherwise 409 HOLD_EXPIRED.
- **Lazy expiry, no scheduler:** every write transaction that locks a flight (book, confirm, cancel) first expires that flight's overdue holds: booking → EXPIRED, seats → RELEASED, counter restored. It runs under the flight lock, so it cannot race with a confirm or a new booking.

- **Reads never write; they compute around overdue holds:** the seat map shows a seat on an overdue hold as `AVAILABLE` ; search adds seats on overdue holds back to the counter (one grouped count query after the instance search); `GET /bookings/{reference}` reports an overdue `HELD` booking as `EXPIRED` .
- **The flag is read only from `AirlineProperties`** : where the `BookingPolicy` is chosen, and in the two places that would otherwise run hold-specific queries (expiry on writes, the overdue-hold count on search). With the flag off those queries are skipped, so the default path pays nothing for the feature; the seat map's single query and `effectiveStatus` need no branch.
- **Seat map states stay two:** a live hold shows as `BOOKED` .

9. Major design decisions

Decision	Chosen	Alternatives considered	Why
Instance generation	Materialised instances, derived seats	Dynamic, lazy hybrid	Stable row to lock and reference; one-query search (\$5)

Concurrency	Flight row lock + partial unique index	Optimistic versioning, index only, JVM locks	Serialises contended writes cheaply, works across nodes, DB backstop (\$6)
Availability	Counter maintained under the lock	<code>COUNT(*)</code> per search	Search needs no aggregation; invariant tested
Cancellation	Soft release (status change)	Delete rows	Keeps history; partial index frees the seat immediately
Booking reference	6-character PNR from an unambiguous alphabet, <code>SecureRandom</code>	UUID, sequence	Readable for customers, not guessable; unique constraint is the final guard
Lookup and cancel	By booking reference only	Reference + surname	No authentication in scope; the reference is the bearer secret
Cancellation rules	One rule ("before departure"), a single check inline in <code>BookingService</code>	A <code>CancellationPolicy</code> Strategy	Only one rule exists; a Strategy earns its place when a second implementation does, as <code>BookingPolicy</code> shows
Seat hold	Strategy behind a flag, lazy expiry	Always-on holds, scheduled clean-up	Brief describes one-step booking; the flag shows the extension without changing default behaviour
Package structure	By feature (<code>schedule</code> , <code>flight</code> , <code>booking</code>), layered inside	Package by layer	Related code changes together; feature boundaries are visible and testable (ArchUnit)
Error format	RFC 9457 <code>ProblemDetail</code> + machine-readable <code>code</code> and <code>requestId</code> on every error, framework errors included	Custom error body	Standard shape, supported natively by Spring; clients branch on <code>code</code> , support traces by <code>requestId</code>
Input-error status	400 for all invalid input; 409 for conflicts with resource state	422 for well-formed but rule-breaking input	Clients branch on <code>code</code> ; one 4xx for "your input is wrong" keeps the contract simple
Database	PostgreSQL 16, Flyway	MySQL, H2	Partial indexes, <code>ON CONFLICT</code> , transactional DDL; tests run on the same engine via Testcontainers
Authentication	None; admin under <code>/admin</code>	Spring Security	Out of scope by the brief; the path split is where role-based access would attach
Time	UTC everywhere; injected <code>Clock</code>	System time	Deterministic tests; no dependence on host time zone

10. Assumptions

From the assignment brief

- The system serves a single airline.
- Airports and aircraft are preloaded through seed data; their CRUD operations are out of scope.
- The seat map is fixed for each aircraft.
- Flight schedules do not change after creation.
- A booking covers a single flight instance, and a seat belongs to only one booking.
- All timestamps are stored in UTC; time-zone conversion is out of scope.
- Payment and refund processing are out of scope.

Additional assumptions

1. **Schedule times:** departure and arrival times entered through the admin API are interpreted as UTC.

2. **Overnight flights:** an arrival time earlier than the departure time means arrival on the following calendar day. Flights longer than 24 hours are out of scope.
3. **Booking window:** flights are bookable from the current UTC date up to and including 365 days ahead; the window advances daily.
4. **Authentication:** user accounts and authentication are out of scope. Administrative endpoints are separated under `/admin` and would be protected by role-based access in production. Bookings are retrieved and cancelled using the booking reference.
5. **Seat hold:** booking is a single step and seats are confirmed immediately, as described in the brief. An optional seat hold can be enabled through configuration (`airline.seat-hold.enabled` , `default false` ; `airline.seat-hold.ttl`). When enabled, seats are held until confirmation or expiry; expired holds are released when availability is next checked. A scheduled clean-up job is not included.
6. **Double booking:** a seat on a flight instance cannot be assigned to more than one active booking. The same passenger name may appear on more than one booking, as passenger validation is not required.
7. **Passengers:** each seat is assigned to one passenger, identified by name only. A booking contains at most 9 passengers.
8. **Departed flights:** flights that have already departed cannot be booked or cancelled.
9. **Cancellation:** cancellation applies to the whole booking; partial cancellation is out of scope. Cancelling an already cancelled booking returns its current state.
10. **Seat map:** the seat map reflects the committed booking state at the time of the request; push updates are out of scope.
11. **Scale:** the design targets a few hundred daily schedules (roughly 100,000 flight instances a year). Application instances are stateless and can be scaled horizontally, with the database as the source of truth for seat inventory.
12. **Flight numbers:** each flight number identifies exactly one schedule.
13. **Schedule validity:** a schedule takes effect from its creation date and has no end date.
14. **Seat selection:** seats are chosen explicitly by the customer; there is no auto-assignment.
15. **Aircraft rotation:** conflicts from one aircraft operating two overlapping flights are not validated.

11. Scaling and evolution

Not built; each step is listed with the signal that would justify it.

Step	Trigger
Horizontal scaling of the service behind a load balancer	Already possible (stateless, DB-enforced correctness); do it when CPU or request latency on one node becomes the limit
Serve search and seat map from a read replica or a short-TTL cache	Read traffic dominates and the primary's load grows; accept slightly stale availability, keep booking on the primary
Partition <code>flight_instance</code> and <code>booking_seat</code> by flight date; archive departed flights	Tables reach tens of millions of rows or index maintenance slows writes
Idempotency key on <code>POST /bookings</code>	Clients retry on timeouts and duplicate bookings appear
Scheduled clean-up of expired holds	Seat hold is enabled and flights with stale holds see few writes, so lazy expiry lags in reports
Finer-grained locking (per seat)	Measured lock waits on a single very popular flight become significant
A <code>CancellationPolicy</code> Strategy (cut-off before departure, admin override, fees), like <code>BookingPolicy</code>	The business defines a second cancellation rule
Partial cancellation (some passengers only)	Customers ask to drop one passenger; seat-level status and <code>released_at</code> already support it, no migration needed
Surname check alongside the reference for lookup and cancel; a mismatch returns 404 like an unknown reference	Self-service is exposed beyond trusted channels without user accounts; a strict match would split <code>passenger_name</code> into first and last name (additive migration)
Value types for seat labels and booking references	Seats gain attributes (class, window/aisle) or seat strings travel through many APIs
Authentication and role-based access on <code>/admin</code>	Any deployment beyond a demo

Cluster-wide lock for the daily job (e.g. ShedLock), so one node runs it	Many nodes each repeating the (idempotent) daily run becomes costly
Distributed tracing (OpenTelemetry via Micrometer Tracing; <code>traceId</code> in the log pattern, <code>traceparent</code> honoured)	A second service or asynchronous messaging. Today correlation is the <code>requestId</code> on every log line (job runs: <code>job-startup-.../job-daily-...</code>) and in every error response
Aircraft-rotation check (same aircraft on overlapping flights, with turnaround time)	Schedules are planned in this system rather than imported from a fleet-planning tool

Multiple airlines

The brief is single-airline, but no part of the design depends on it.

- **Already works with no change.** The airline code is the first two characters of every flight number (`XY101` , `EK001`), so schedules of several airlines coexist and their flight numbers cannot clash under the existing unique constraint. Flights of different airlines on the same route at the same time are normal and need no validation; search returns all of them. The flight lock is per flight instance, so airlines never block each other. Instance generation, booking, cancellation and the seat hold are unchanged.
- **Aircraft ownership and airline data** → an `airline` reference table (like `airport`) and `aircraft.airline_code` (one additive migration), plus a check in `ScheduleService` that a schedule's flight-number prefix matches its aircraft's airline. Today any schedule may use any aircraft, and the prefix is not validated. *Trigger: a second airline is onboarded.*
- **Codeshares** → store the operating carrier explicitly; the flight-number prefix is only the marketing carrier. *Trigger: flights sold under another airline's number.*
- **Isolation between airlines** → an airline id on owned tables enforced on every query (e.g. PostgreSQL row-level security) and airline-scoped admin authentication. *Trigger: the system is offered as a platform to separate airlines.*
- **Not covered for one airline or many:** aircraft-rotation clashes (table above) and airport slot or capacity limits (handled by slot coordination outside a booking system).

Changelog

Date	Change
2026-10-07	Initial design
2026-10-08	Phase 2: <code>ApiException</code> is one concrete class carrying an <code>ErrorCode</code> ; subclasses only for errors with extra fields (LLD §2)
2026-10-08	Phase 3: search adds back seats on overdue holds with a second grouped query, not a correlated subquery (HLD §8, LLD §8)
2026-10-08	Phase 4: the booking seat-conflict check reuses <code>SeatOccupancyQueries.takenSeats</code> , so seat map, search and booking share one definition of a taken seat (LLD §2, §7)
2026-10-08	Phase 5: cancellation rules behind <code>CancellationPolicy</code> ; lock order and idempotent cancel as designed (no change to this document)
2026-10-08	Phase 4b: <code>hold_expires_at</code> is kept after a hold ends (ER diagram and schema snapshot wording); hold expiry rounded to whole seconds; expiry flushes before the SQL seat check (LLD §8)
2026-10-08	Phase 6: incoming <code>X-Request-Id</code> accepted only if safe (≤ 64 chars, <code>[A-Za-z0-9._-]</code>), else a UUID (LLD §2)
2026-10-08	Configurable window: <code>airline.booking-window-days</code> is the single definition of the window (<code>AirlineProperties.lastBookableDate</code>), now also enforced at booking; <code>Retry-After</code> is configuration (<code>airline.retry-after</code>); Compose passes <code>AIRLINE_*</code> overrides through (LLD §2, §5, §6)

2026-10-08	Phase 8: documentation finished: links to README and ADRs; multiple-airlines and aircraft-rotation notes in §11; LLD aligned with the final code
2026-10-08	Phase 9: manual top-up endpoint <code>POST /admin/instance-window/extend</code> (§5); a booking-reference clash returns 503 <code>RETRY_LATER</code> instead of 500; <code>airline.retry-after</code> shared by both 503s; job runs log under their own id; tracing and scheduler-lock rows in §11 (LLD §2, §4, §6, §7, §9)
2026-10-09	Phase 10 (hardening): cancellation rule inlined (no <code>CancellationPolicy</code>); hold queries skipped with the flag off (§8); instances generated one day past the window (§5); startup top-up never aborts boot; V5 composite FK and format CHECKs (data model); PostgreSQL published on 5433